Towards Formalizing Reinforcement Learning Theory: A Robbins-Siegmund Approach

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Summary

In this paper, we formalize the almost sure convergence of Q-learning and linear temporal difference (TD) learning with Markovian samples using the Lean 4 theorem prover based on the Mathlib library. Q-learning and linear TD are among the earliest and most influential reinforcement learning (RL) algorithms. The investigation of their convergence properties is not only a major research topic during the early development of the RL field but also receives significant attention nowadays. This paper formally verifies their almost sure convergence in a unified framework based on the Robbins-Siegmund theorem. The framework developed in this work can potentially be extended to convergence rates and other modes of convergence. This work thus makes an important step towards fully formalizing convergent RL results.

Contribution(s)

1. The first formal verification of the almost sure convergence of Q-learning and linear TD with Markovian samples.
   Context: Some results of dynamic programming and RL are formalized in existing works (Vajjha et al., 2021; 2022; Chevallier & Fleuriot, 2021; Schäfeller & Abdulaziz, 2022; Chevallier, 2024; Schäffeler & Abdulaziz, 2025). But those works are far from formally verifying almost sure convergence of Q-learning and linear TD with Markovian samples. Some recent works (Li et al., 2024a; 2025a;b; Sonoda et al., 2025) formalize generalization bounds (via Rademacher complexity) and optimality of a few first-order optimization methods. They are also far from RL.
2. The formalization of a reusable Lean infrastructure based on the Robbins-Siegmund theorem for convergence of RL algorithms, instantiated here for linear TD and Q-learning to show effectiveness.
   Context: Historically, the almost sure convergence analysis of RL algorithms is focused on ODE-based approaches (Benveniste et al., 1990; Kushner & Yin, 2003; Borkar, 2009; Liu et al., 2025a). However, the current infrastructure in Mathlib does not support an easy formalization of ODE-based approaches. Instead, this work formalizes a more recent framework (Qian et al., 2024) based on the Robbins-Siegmund theorem. We demonstrate the effectiveness of this framework by using this framework to formalize the almost sure convergence of Q-learning and linear TD with Markovian samples. We argue that this framework is highly extensible because this framework has been used to provide asymptotic and nonasymptotic analyses of many RL algorithms in many different modes (e.g., almost surely, with high probability, in $\mathcal{L}^p$) (Qian et al., 2024; Liu et al., 2025b). We do note that those perceived extensions are not formalized yet and here we just argue for the extensibility.

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Abstract

In this paper, we formalize the almost sure convergence of Q-learning and linear temporal difference (TD) learning with Markovian samples using the Lean 4 theorem prover based on the Mathlib library. Q-learning and linear TD are among the earliest and most influential reinforcement learning (RL) algorithms. The investigation of their convergence properties is not only a major research topic during the early development of the RL field but also receives significant attention nowadays. This paper formally verifies their almost sure convergence in a unified framework based on the Robbins-Siegmund theorem. The framework developed in this work can potentially be extended to convergence rates and other modes of convergence. This work thus makes an important step towards fully formalizing convergent RL results. The code is available at https://github.com/ShangtongZhang/rl-theory-in-lean.

1 Introduction

Narrowly speaking, reinforcement learning (RL, Sutton & Barto (2018)) is a framework for solving sequential decision making problems via trial and error. Q-learning (Watkins, 1989; Watkins & Dayan, 1992) and linear temporal difference (TD) learning (Sutton, 1988) are among the earliest and most influential RL algorithms. The investigation of their convergence property constitutes an important research topic in the RL theory community (Watkins, 1989; Watkins & Dayan, 1992; Dayan, 1992; Jaakkola et al., 1993; Tsitsiklis, 1994; Tsitsiklis & Van Roy, 1997; Kearns & Singh, 1998; Tsitsiklis & Roy, 1999; Even-Dar et al., 2003; Azar et al., 2011; Beck & Srikant, 2012; Shah & Xie, 2018; Bhandari et al., 2018; Lakshminarayanan & Szepesvári, 2018; Srikant & Ying, 2019; Lee & He, 2020; Qu & Wierman, 2020; Li et al., 2020; Chen et al., 2024; Li et al., 2024b; Meyn, 2024; Liu et al., 2025c; Xie et al., 2025; Liu et al., 2025b; Blaser et al., 2026; Blaser & Zhang, 2026; Wang & Zhang, 2026).

We, however, argue that RL convergence proofs are usually delicate for two reasons. **First**, the almost sure convergence of RL algorithms is usually established through the ODE approach (Benveniste et al., 1990; Kushner & Yin, 2003; Borkar, 2009; Borkar et al., 2025; Liu et al., 2025a). For example, the seminal work Tsitsiklis & Van Roy (1997) that establishes the almost sure convergence of linear TD relies on an ODE based stochastic approximation result in Benveniste et al. (1990). The ODE based approach is full of details and bug-prone. For example, Degris et al. (2012) investigate the convergence of off-policy actor critic algorithms through an ODE based approach, but as suggested by their erratum, one major result in their peer-reviewed accepted version is entirely wrong. Wan et al. (2021) investigate the convergence of average reward RL algorithms and point out that an earlier peer-reviewed accepted work gives a pseudoproof of the major result of Wan et al. (2021). Even a well-established textbook makes gaps. For example, the second version of Borkar (2009) states (and fixes) a major gap in its first version. Those are publicly documented gaps (with or without fixes), with more gaps hidden and only known to experts as folklore. **Second**, RL theory

is typically formulated in the framework of Markov Decision Process (MDP, Bellman (1957); Puterman (1994)). To rigorously study the convergence of stochastic iterates inside the MDP framework, one has to first construct the probability space for infinite length trajectories of the MDP (i.e., sample paths). This inevitably requires using the Ionescu-Tulcea theorem (Tulcea, 1949). Consequently, one has to verify the measurability and integrability of many functions in this probability space. One also has to use a measure theoretic definition of conditional expectations with sub-σ-algebras in this probability space. To reduce this definition to a more easy-to-use plain definition with conditional probability, one again needs to verify the measurability and integrability of many related functions. To our knowledge, no prior RL theory work has gone through all those details to give a full rigor of their results. Likely, for finite state action MDPs, those measurability and integrability can eventually be verified. But for infinite ones, there is a good chance that more assumptions are needed in existing results. We regard formal verification as the ultimate approach for robustifying RL theory. Accordingly, this work develops the first formalization of the almost sure convergence of Q-learning and linear TD with Markovian samples on finite state action MDPs, using the Lean 4 theorem prover (de Moura & Ullrich, 2021) based on the Mathlib library (The mathlib Community, 2020).[1]

Work has been done to formalize the basics of RL. However, those are better categorized as formalizing dynamic programming instead of RL as there is no stochasticity in the algorithms they consider. For example, Vajjha et al. (2021); Chevallier & Fleuriot (2021); Schäfeller & Abdulaziz (2022); Schäffeler & Abdulaziz (2025) formalize the optimality of a few dynamic programming algorithms, e.g., (approximate) policy and value iteration, in Coq or Isabelle/HOL. More related are Vajjha et al. (2022); Chevallier (2024), which formalize Dvoretzky's theorem (Dvoretzky, 1955) for the almost sure convergence of a class of stochastic approximation algorithms in Coq and Isabelle/HOL, respectively. Dvoretzky's theorem can be used to prove the almost sure convergence of some (arguably outdated) version of Q-learning (more details in Section 2), but such a proof is never formalized in any prior work and Dvoretzky's theorem is incapable of analyzing linear TD. By contrast, this paper is centered around modern techniques combining Lyapunov functions (Chen et al., 2024) and the Robbins-Siegmund theorem (Robbins & Siegmund, 1971) by using a skeleton iterates technique (Qian et al., 2024) to convert Markovian noise to Martingale difference noise, which provides a unified framework for formalizing not only almost sure convergence but also high probability concentration, $\mathcal{L}^p$ convergence, and the corresponding convergence rates. Notable fellow projects within the machine learning community includes FoML (Sonoda et al., 2025) and Optlib (Li et al., 2024a; 2025a;b), both of which are in Lean. FoML formalizes generalization bound by Rademacher complexity and is distant from the convergence of RL algorithms. Optlib formalizes the optimality of a few first-order optimization methods (e.g., ADMM, gradient descent, Nesterov's accelerated methods). While first-order optimization methods are closer to RL algorithms, Optlib does not have any stochasticity either. To summarize, the focus on RL algorithms with Markov chain driven stochasticity distinguishes this paper from prior works of this kind.

The formalization can also serve as a high quality dataset for benchmarking LLM's reasoning and coding capability. One example is Yang et al. (2025), which develop a pipeline (centered around a new tactic in Lean) that can generate many subgoals from a complete Lean proof. Those subgoal completion problems can then be used to benchmark LLMs.

LLM statements. The formalization done in this paper greatly benefits from LLMs (specifically, Gemini and ChatGPT) in three ways. First, LLM serves as a personalized tutor that significantly bends the notoriously sharp learning curve of Lean. Second, LLM serves as a powerful search engine that can effectively retrieve corresponding lemmas from Mathlib based on natural language descriptions. Third, LLM can complete some small lemmas automatically in one trial. The autocompletion powered by Copilot is also very helpful in refactoring the implementation. In this sense, this project can be regarded as a collaboration between humans and AI where humans dominate the collaboration. With the help of LLMs, we are able to complete this project in three months in part-time starting with zero knowledge of Lean. A natural question for better gauging the contribution of this work is then: *can LLM complete this project alone or with little help from humans?*

[1] For readers unfamiliar with Lean and Mathlib, a brief introduction is available in Appendix A.

Our answer is negative as of Nov 2025. First, from our own experience of interacting with LLMs during the project, we frequently see hallucinations and significant incapacities of LLMs. More importantly, LLMs are mostly good at proving theorem statements that are already formalized and such proofs can be checked automatically. However, we believe reliably and precisely converting theorem statements from natural language to formal statements and designing an extensible framework is still beyond the reach of current LLMs. Second, the recent Gauss agent (Math-Inc, 2025) achieves an important milestone in automated formalization by completely formalizing the strong prime number theorem. In addition to the huge amount of computation the Gauss agent consumes, it still receives significant input from humans in two ways. First, the Gauss agent does not start from scratch. Instead, it starts from some important milestones (towards formalizing the strong prime number theorem) made by humans. Second, the Gauss agent relies on an 83-page human-written blueprint as a roadmap for formalization. This blueprint is iterated multiple times by humans based on the progress and failures the agent makes. Part of the blueprint is very detailed. For example, it can contain trivial lemmas such as the absolute value of a positive real is itself.

2 Background

Notations. For $x, y \in \mathbb{R}^d$, we use $\langle \cdot, \cdot \rangle$ to denote the inner product in Euclidean space, i.e., $\langle x, y \rangle = x^\top y$. We use $\|x\|_p \doteq \left(\sum_i |x_i|^p\right)^{1/p}$ to denote the ℓ_p norm and $\|x\|_\infty \doteq \max_i |x_i|$ to denote the infinity norm. We overload the vector norms to also denote the induced matrix norms.

We consider an infinite horizon MDP with a finite state space $\mathcal{S}$, a finite action space $\mathcal{A}$, a reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, a transition function $p : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \to [0, 1]$, an initial distribution $p_0 : \mathcal{S} \to [0, 1]$, and a discount factor $\gamma \in [0, 1)$. At time step 0, an initial state S_0 is sampled from p_0. At time t and state S_t, an action A_t is sampled from $\pi(\cdot|S_t)$, where $\pi : \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the policy. A successor state S_{t+1} is then sampled from $p(\cdot|S_t, A_t)$ and a reward $R_{t+1} \doteq r(S_t, A_t)$ is generated. The state value function is defined as $v_\pi(s) \doteq \mathbb{E}\left[\sum_{i=0}^\infty \gamma^i R_{t+i+1} | S_t = s\right]$ and the action value function is defined as $q_\pi(s, a) \doteq \mathbb{E}\left[\sum_{i=0}^\infty \gamma^i R_{t+i+1} | S_t = s, A_t = a\right]$. Estimating v_π is one fundamental task in RL, called policy evaluation. Another fundamental task is control, the goal of which is to find an optimal policy π_* such that $q_{\pi_*}(s, a) \geq q_\pi(s, a)\ \forall \pi, s, a$. There can be multiple optimal policies but all of them must share the same action value function, denoted as q_* and called the optimal action value function, which is the unique fixed point of the Bellman optimality operator $\mathcal{T}_* : \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|} \to \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ defined as $(\mathcal{T}_* q)(s, a) \doteq r(s, a) + \gamma \sum_{s'} p(s'|s, a) \max_{a'} q(s', a')$.

TD is one of the most well-received algorithms for policy evaluation, which estimates v_π via stochastic iterates $\left\{v_t \in \mathbb{R}^{|\mathcal{S}|}\right\}$ generated as $v_{t+1}(s) = v_t(s) + \alpha_t(R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))\mathbb{I}_{s=S_t}$, where $\{\alpha_t\}$ is a sequence of deterministic step sizes and $\mathbb{I}$ is the indicator function. Q-learning is one of the most well-received algorithms for control, which estimates q_* via stochastic iterates $\left\{q_t \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}\right\}$ generated as

$$q_{t+1}(s, a) = q_t(s, a) + \alpha_t(R_{t+1} + \gamma \max_b q_t(S_{t+1}, b) - q_t(S_t, A_t))\mathbb{I}_{(s,a)=(S_t,A_t)}. \quad (Q\text{-learning})$$

Here we consider the case where the action A_t is sampled from a fixed policy π. It is well-known (e.g., Qian et al. (2024)) that almost surely, $\lim_{t\to\infty} v_t = v_\pi$ and $\lim_{t\to\infty} q_t = q_*$. Instead of using a look-up table $\{v_t\}$ to store estimates of v_π, parameterized functions can also be used. Particularly, Sutton (1988) considers a linear parameterization. Let $x : \mathcal{S} \to \mathbb{R}^K$ be a feature function that maps a state s to a K-dimensional feature. We then use $x(s)^\top w$ to approximate $v_\pi(s)$, where $w \in \mathbb{R}^K$ is a learnable weight. Linear TD then generates iterates $\left\{w_t \in \mathbb{R}^K\right\}$ as

$$w_{t+1} = w_t + \alpha_t(R_{t+1} + \gamma x(S_{t+1})^\top w_t - x(S_t)^\top w_t)x(S_t). \quad (\text{Linear TD})$$

It is well-known (e.g., Tsitsiklis & Van Roy (1997)) that $\lim_{t\to\infty} w_t = w_*$ a.s., where w_* is the TD fixed point. To define w_*, we use $X \in \mathbb{R}^{|\mathcal{S}| \times K}$ to denote the feature matrix whose s-th row is $x(s)^\top$, use $P_\pi \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ to denote the transition matrix of the Markov chain induced by π such that $P_\pi(s, s') = \sum_a \pi(a|s)p(s'|s, a)$, use $r_\pi \in \mathbb{R}^{|\mathcal{S}|}$ to denote the reward vector such that $r_\pi(s) =$

$\sum_a \pi(a|s) r(s,a)$, and use $D_\pi \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}$ to denote the diagonal matrix whose diagonal term is the stationary state distribution $d_\pi \in \mathbb{R}^{|\mathcal{S}|}$ of the Markov chain $\{S_t\}$ induced by π. Then we have $w_* = -A^{-1}b$, where $A \doteq X^\top D_\pi(\gamma P_\pi - I)X$ and $b \doteq X^\top D_\pi r_\pi$.

The goal of this paper is thus to formally prove that $\lim_{t\to\infty} q_t = q_*$ a.s. and $\lim_{t\to\infty} w_t = w_*$ a.s. We note that for (Linear TD), we follow Tsitsiklis & Van Roy (1997) and consider a Markov Reward Process (MRP) setup, i.e., we use $R_{t+1} \doteq r_\pi(S_t)$ directly in (Linear TD). We also note that earlier works of Q-learning use a different update rule that replaces α_t in (Q-learning) with $\alpha_{\nu(S_t,A_t,t)}$, where $\nu(s,a,t) \doteq \sum_{\tau=0}^{t} \mathbb{I}_{(s,a)=(S_i,A_i)}$ is a counter that counts the visit of (s,a) up to time t. This counter is rarely used by practitioners or the modern formulation of Q-learning (Sutton & Barto, 2018) and Antrobius & Zhang (2026) show that such a counter is mostly a technical convenience to significantly simplify the proof. We, therefore, do not consider this counter in our formalization. The form (Q-learning) is also what Chevallier (2024) proposed to formalize (but did not) after they formalized Dvoretzky's theorem. Chevallier (2024) states that the formal proof of (Q-learning) is very close after the formal proof of Dvoretzky's theorem. We, however, argue that there is a gap. If the counter-based step size was used in (Q-learning), then the convergence would follow easily from Dvoretzky's theorem. But for the exact form of (Q-learning), one needs to additionally prove that $\forall(s,a), \sum_{t=0}^{\infty} \alpha_t \mathbb{I}_{(s,a)=(S_t,A_t)} = \infty$ a.s. to use Dvoretzky's theorem. This is true under moderate assumptions on the Markov chain but highly nontrivial to formalize, especially given that Chevallier (2024) does not have a measure theoretic formalization of the probability space of sample paths of the Markov chain.

3 Formal Theorem Statements

We now describe our formalization of the theorem statements. We start with the almost sure convergence of (Linear TD). To this end, we first define stochastic vectors on a finite state space $\mathcal{S}$ and the corresponding row stochastic matrix.

```
variable {S : Type u} [Fintype S]
class StochasticVec (x : S → ℝ) where
  nonneg : ∀ s, 0 ≤ x s
  rowsum : ∑ s, x s = 1
class RowStochastic (P : Matrix S S ℝ) where
  stochastic: ∀ s, StochasticVec (P s)
```

We then define irreducibility and aperiodicity of row stochastic matrices.

```
class Irreducible (P : Matrix S S ℝ) [RowStochastic P] where
  irreducible : ∀ i j, ∃ n : ℕ, 0 < (P ^ n) i j
class Aperiodic (P : Matrix S S ℝ) [RowStochastic P] where
  aperiodic : ∀ i, FiniteGCDOne (return_times P i)
```

Irreducibility and aperiodicity together imply Doeblin minorization after sufficient powers.

```
class DoeblinMinorization (P : Matrix S S ℝ) [RowStochastic P] where
  minorize : ∃ (ϵ : ℝ) (ν : S → ℝ), 0 < ϵ ∧ ϵ < 1 ∧ StochasticVec ν ∧ ∀ i j, P i j ≥ ϵ * ν j
theorem smat_minorizable_with_large_pow [Nonempty S] (P : Matrix S S ℝ)
  [RowStochastic P] [Irreducible P] [Aperiodic P] : ∃ N, 1 ≤ N ∧ DoeblinMinorization (P ^ N)
```

The corresponding matrix power operator is then a contraction in the simplex.

```
theorem smat_contraction_in_simplex (P : Matrix S S ℝ) [RowStochastic P]
  [DoeblinMinorization P] : ∃ K, 0 < K ∧ ContractingWith K (smat_as_operator P)
```

This allows us to invoke Banach's fixed point theorem to conclude the existence and uniqueness of the stationary distribution as well as the geometric mixing property.

```
theorem stationary_distribution_uniquely_exists (P : Matrix S S ℝ) [RowStochastic P]
  [Aperiodic P] [Irreducible P] : ∃! μ : S → ℝ, StochasticVec μ ∧ Stationary μ P
instance (P : Matrix S S ℝ) [RowStochastic P] [Aperiodic P] [Irreducible P]
  : GeometricMixing P
```

Having defined the stationary distribution, we are finally ready to formalize the TD fixed point w_*.

```
abbrev E (d : ℕ) := EuclideanSpace ℝ (Fin d)
noncomputable def LinearTDSpec.td_fixed_point : E d := - spec.A⁻¹ *ᵥ spec.b
```

We now describe how we construct the sample path probability space. To this end, we first define a time-homogeneous Markov chain using probability kernels from Mathlib.

```
structure HomMarkovChainSpec (S : Type u) [MeasurableSpace S] where
  kernel : Kernel S S
  markov_kernel : IsMarkovKernel kernel
  init : ProbabilityMeasure S
```

Notably, in Mathlib, `IsMarkovKernel` only means the kernel is a probability kernel and has nothing to do with the Markovian property of a Markov chain. We then generate the probability measure on the sample path space S^∞.

```
noncomputable def traj_prob (M : HomMarkovChainSpec S) : ProbabilityMeasure (ℕ → S)
```

This is done by invoking the Ionescu-Tulcea theorem in Mathlib (Marion, 2025), which uses a constructive way to prove the existence of a probability measure on S^∞ that coincides with the iterated application of the transition kernel P on any partial sample path with finite length. The best practice (e.g., Tsitsiklis & Van Roy (1997)) for analyzing the convergence of linear TD in existing literature is to consider the augmented Markov $Y_t \doteq (S_t, S_{t+1})$. We follow this and eventually realize `HomMarkovChainSpec` with a state space $\mathcal{Y} \doteq \mathcal{S} \times \mathcal{S}$. The corresponding transition kernel, described here in a matrix form for simplicity, is then $P_{\mathcal{Y}}((s_0, s_0'), (s_1, s_1')) = \mathbb{I}_{s_1 = s_0'} P_\pi(s_1, s_1')$. The Ionescu-Tulcea theorem then generates a probability measure on $(S \times S)^\infty$. Given a sample path $\omega \in (S \times S)^\infty$, the algorithm (Linear TD) is then defined as

```
noncomputable def LinearTDSpec.update (w : E d) (y : S × S) : E d :=
  (spec.r y.1 + spec.γ * ⟪spec.x y.2, w⟫ - ⟪spec.x y.1, w⟫) • spec.x y.1
variable {w : ℕ → (ℕ → (S × S)) → E d}
class LinearTDIterates where
  init : ∀ ω, w 0 ω = spec.w₀
  step : ∀ n ω, w (n + 1) ω = w n ω + spec.α n • spec.update (w n ω) (ω (n + 1))
```

We now formalize the almost sure convergence of linear TD as

```
theorem ae_tendsto_of_linearTD_markov
  {ν : ℝ} (hν : ν ∈ Set.Ioo (2 / 3) 1) (hw : LinearTDIterates (spec := spec) (w := w))
  (hα : spec.α = fun n : ℕ => inv_poly ν 2 n) :
  ∀ᵐ ω ∂ spec.markov_samples, Tendsto (fun n => w n ω) atTop (𝓝 spec.td_fixed_point)
```

Precisely, what we formalize is

Theorem 3.1 *Let the finite Markov chain $\{S_t\}$ be irreducible and aperiodic. Let X have a full column rank. Let the step size be $\alpha_t = \frac{1}{(t+2)^\nu}$ with $\nu \in (2/3, 1)$. Then the iterates $\{w_t\}$ generated by* (Linear TD) *with $R_{t+1} \doteq r_\pi(S_t)$ satisfy that $\lim_{t\to\infty} w_t = w_*$ a.s.*

The particular choice of $\nu \in (2/3, 1)$ is an artifact of our proof technique and we shall revisit this in the next section. We similarly formalize the almost sure convergence of Q-learning as

```
theorem ae_tendsto_of_QLearning_markov
  {ν : ℝ} (hν : ν ∈ Set.Ioo (2 / 3) 1) (hq : QLearningIterates (spec := spec) (q := q))
  (hα : spec.α = fun n : ℕ => inv_poly ν 2 n) :
  ∀ᵐ ω ∂ spec.MRP.markov_samples, Tendsto (fun n => q n ω) atTop (𝓝 spec.optimal_q)
```

Precisely, what we formalize is

Theorem 3.2 *For any fixed policy π, let the induced finite Markov chain $\{(S_t, A_t)\}$ be irreducible and aperiodic. Let the step size be $\alpha_t = \frac{1}{(t+2)^\nu}$ with $\nu \in (2/3, 1)$. Then the iterates $\{q_t\}$ generated by* (Q-learning) *satisfy that $\lim_{t\to\infty} q_t = q_*$ a.s.*

We also formalize the two almost sure convergences under i.i.d. samples, where more step sizes are allowed as long as the step sizes satisfy the Robbins-Monro condition[2].

4 Formal Theorem Proofs

The canonical almost sure convergence analysis of linear TD relies on ODE-based methods (Benveniste et al., 1990; Kushner & Yin, 2003; Borkar, 2009; Borkar et al., 2025; Liu et al., 2025a). How-

[2]In this work, we say a sequence $\{\alpha_t\}$ satisfies the Robbins-Monro condition if $0 < \alpha_t, \sum_t \alpha_t = \infty, \sum_t \alpha_t^2 < \infty$.

ever, our evaluation is that those ODE-based approaches are not ready for formalization yet. The main reason is that Mathlib has only very few results about ODE and control theory. Instead, this paper uses a more modern approach based on the Robbins-Siegmund theorem. We now elaborate more on our roadmap, which is largely based on Chen et al. (2024); Qian et al. (2024). Both (Q-learning) and (Linear TD) can be rewritten in the form of

$$w_{t+1} = w_t + \alpha_t(F(w_t, Y_{t+1}) - w_t). \tag{1}$$

For (Linear TD), we have $Y_{t+1} \doteq (S_t, S_{t+1})$ and $F(w,(s,s')) = (r_\pi(s)+\gamma x(s')^\top w - x(s)^\top w)x(s) + w$. For ($Q$-learning), we have $Y_{t+1} \doteq (S_t, A_t, S_{t+1}, A_{t+1})$ and $F(q,(s,a,s',a'))(s_0,a_0) = (r(s,a) + \gamma \max_b q(s',b) - q(s,a))\mathbb{I}_{(s,a)=(s_0,a_0)} + q(s_0,a_0)$. Notably, F actually does not depend on the argument a'. It is included here only for a unified proof implementation for both (Q-learning) and (Linear TD). Let $f(w)$ be the expectation of $F(w,\cdot)$ w.r.t. the stationary distribution. We then have

$$w_{t+1} = w_t + \alpha_t(f(w_t) - w_t) + \alpha_t(F(w_t, Y_{t+1}) - f(w_t)). \tag{2}$$

This motivates us to study the iterates described below.

```
class Iterates (x : ℕ → ω → E d)
  (x₀ : E d) (f : E d → E d) (e₁ e₂ : ℕ → ω → E d) (α : ℕ → ℝ) where
  init : ∀ ω, x 0 ω = x₀
  step : ∀ n ω, x (n + 1) ω =
    x n ω + (α n) • (f (x n ω) - x n ω) + e₁ (n + 1) ω + e₂ (n + 1) ω
```

Precisely, let Ω be a set equipped with a σ-algebra, let μ be a probability measure, and let $\{x_n, e_{1,n}, e_{2,n} : \Omega \to \mathbb{R}^d\}$ be a sequence of measurable functions satisfying $\forall \omega \in \Omega$

$$x_{n+1}(\omega) = x_n(\omega) + \alpha_n(f(x_n(\omega)) - x_n(\omega)) + e_{1,n+1}(\omega) + e_{2,n+1}(\omega). \tag{3}$$

We then study the convergence of x_n under assumptions on the noise terms $e_{1,n}$ and $e_{2,n}$. Here $f : \mathbb{R}^d \to \mathbb{R}^d$ is the function of interest that has a fixed point x_* such that $f(x_*) = x_*$. Our goal is thus to show $\lim_{n\to\infty} x_n(\omega) = x_*$ for almost all ω. We make some basic assumptions on f and α_n.

Assumption 4.1 *$\{\alpha_n\}$ satisfy the Robbins-Monro condition and f is Lipschitz continuous.*

We further assume the existence of a Lyapunov function $\phi : \mathbb{R}^d \to [0,\infty)$ that satisfies

Assumption 4.2 $\forall x, y$

(a) $\phi(y) \le \phi(x) + \langle \nabla\phi(x), y - x\rangle + C\|y - x\|_2^2$

(b) $\phi(x) \ge 0$ *and* $\phi(x) = 0 \iff x = 0$

(c) $\langle \nabla\phi(x), x\rangle = C\phi(x), \sum_i |(\nabla\phi(x))_i||y_i| \le C\sqrt{\phi(x)}\sqrt{\phi(y)}, \|x\|_2 \le C\sqrt{\phi(x)}, \sqrt{\phi(x)} \le C\|x\|_2$

(d) $\langle \nabla\phi(x - x_*), f(x) - x\rangle \le -\eta\phi(x - x_*)$

Here C just denotes the existence of some nonnegative constants and C does not need to be the same for each of its appearances. Notably, η needs to be strictly positive. Assumption 4.2 (a) essentially says that ϕ is smooth. Assumptions 4.2 (b) & (c) says that ϕ needs to behave like a squared norm. Assumption 4.2 (d) says the update direction $f(x) - x$ should decay the Lyapunov function. We prove that $\phi(x) = \frac{1}{2}\|x\|_p^2$ satisfies (a), (b), and (c) for $p \ge 2$. When $p = 2$, (d) is verified for the f corresponding to (Linear TD). For linear TD, it can be computed that $f(w) = Aw + b + w$. Then

$$\left\langle \nabla \tfrac{1}{2}\|w - w_*\|_2^2, f(w) - w \right\rangle = \langle w - w_*, Aw + b\rangle = \langle w - w_*, A(w - w_*)\rangle \le -\eta\|w - w_*\|_2^2,$$

where the second equality is due to $Aw_* + b = 0$ and the last inequality is due to that A is negative definite. When p is sufficiently large, (d) is verified for the f corresponding to (Q-learning). Specifically, for (Q-learning), define a weighted Bellman optimality operator as $(\mathcal{T}'_* q)(s,a) \doteq d_{\pi_q}(s)\pi_q(a|s)[(\mathcal{T}_* q)(s,a) - q(s,a)] + q(s,a)$. For this weighted Bellman optimality operator, the

behavior policy π is allowed to depend on the action value estimation q. Liu et al. (2025c) prove that $\mathcal{T}'_*$ is a pseudo-contraction, i.e., there exists a $\gamma' \in [0,1)$ such that $\forall q, \|\mathcal{T}'_* q - q_*\|_\infty \le \gamma' \|q - q_*\|_\infty$. In this paper, we consider the setup where the behavior policy is fixed so π_q degenerates to π directly. For our f corresponding to (Q-learning), it can be computed that $f(q) = \mathcal{T}'_* q$. We then have (see Section B in the supplementary materials for a brief proof)

$$\left\langle \nabla \tfrac{1}{2}\|q - q_*\|_p^2, f(q) - q \right\rangle \le -(1 - \gamma'(|\mathcal{S}||\mathcal{A}|)^{1/p})\|q - q_*\|_p^2. \tag{4}$$

For sufficiently large p, we have $\eta \doteq (1-\gamma'(|\mathcal{S}||\mathcal{A}|)^{1/p}) > 0$. Back to (3), we now make assumptions on the growth of the noise terms.

Assumption 4.3 *There exists $C \ge 0$ such that $\forall n$ and almost every ω,*

$$\|e_{1,n+1}(\omega)\|_2 \le C\alpha_n(1 + \|x_n(\omega)\|^2), \ \|e_{2,n+1}(\omega)\|_2 \le C\alpha_n^2(1 + \|x_n(\omega)\|^2).$$

We are now able to prove a recursive error bound for almost every ω.

Lemma 4.4 *Let Assumptions 4.1 - 4.3 hold. Then there exist some constants $C_1 > 0, C_2 \ge 0$, and $n_0 \ge 0$ such that $\forall n \ge n_0$ and for almost every ω*

$$\phi(x_{n+1}(\omega) - x_*) \le (1 - C_1\alpha_n)\phi(x_n(\omega) - x_*) + \langle \nabla\phi(x_n(\omega) - x_*), e_{1,n+1}(\omega)\rangle + C_2\alpha_n^2. \tag{5}$$

We now further assume that $\{e_{1,n}\}$ is a Martingale difference sequence.

Assumption 4.5 *$\{x_n\}$ is adapted to a filtration $\{\mathcal{F}_n\}$ and $\mathbb{E}[e_{1,n+1}|\mathcal{F}_n] = 0$ a.s.*

Here we recall that the conditional expectation $\mathbb{E}[e_{1,n+1}|\mathcal{F}_n]$ is the unique (up to nullset of μ) function $\Omega \to \mathbb{R}^d$ such that for any $B \in \mathcal{F}_n, \int_B \mathbb{E}[e_{1,n+1}|\mathcal{F}_n]\mathrm{d}\mu = \int_B e_{1,n+1}\mathrm{d}\mu$. Taking conditional expectations on both sides of (5) then generates that

$$\mathbb{E}[\phi(x_{n+1}(\omega) - x_*)|\mathcal{F}_n] \le (1 - C_1\alpha_n)\mathbb{E}[\phi(x_n(\omega) - x_*)|\mathcal{F}_n] + C_2\alpha_n^2 \quad \text{a.s.}$$

This means that the sequence of functions $\{\omega \mapsto \phi(x_n(\omega) - x_*)\}$ is almost a supermartingale. By a special case of the Robbins-Siegmund theorem formalized below,

```
theorem ae_tendsto_zero_of_almost_supermartingale
  (hAdapt : Adapted 𝓕 f) (hfm : ∀ n, Measurable (f n)) (hfInt : ∀ n, Integrable (f n) μ)
  (hfnonneg : ∀ n, 0 ≤ᵐ[μ] f n) {T : ℕ → ℝ} (hTpos  : ∀ n, 0 < T n)
  {hTsum : Tendsto (fun n => ∑ k ∈ range n, T k) atTop atTop}
  {hTsqsum : Summable (fun n => (T n) ^ 2)}
  (hAlmostSupermartingale : ∃ C ≥ 0, ∀ n,
    μ[f (n + 1) | 𝓕 n] ≤ᵐ[μ] (fun ω => (1 - T n) * f n ω + C * T n ^ 2)) :
  ∀ᵐ ω ∂μ, Tendsto (fun n => f n ω) atTop (𝓝 0)
```

we obtain that $\lim_{n\to\infty} \phi(x_n(\omega) - x_*) = 0$ a.s. Precisely, the version of the Robbins-Siegmund theorem and the stochastic approximation result we formalize so far are

Theorem 4.6 (A special case of Robbins & Siegmund (1971)) *Let $\{z_n : \Omega \to \mathbb{R}\}$ be a sequence of functions such that $z_n \ge 0$ a.s. and z_n is integrable. Let $\{\mathcal{F}_n\}$ be a filtration such that z_n is measurable by $\mathcal{F}_n$. Let $\{T_n\}$ be a sequence of deterministic reals satisfying the Robbins-Monro condition. If $\{z_n\}$ is almost a supermartingale given $\{T_n\}$ and some nonnegative constant C in the sense that $\mathbb{E}[z_{n+1}|\mathcal{F}_n] \le (1 - T_n)z_n + CT_n^2$ a.s., then $\lim_{n\to\infty} z_n = 0$ a.s.*

Theorem 4.7 *Let Assumptions 4.1 - 4.5 hold. Then $\{x_n(\omega)\}$ in (3) satisfy $\lim_{n\to\infty} x_n(\omega) = x_*$ a.s.*

Looking back at (3), the roles of the two noise terms are clearer now. The noise e_1 is larger (of $\mathcal{O}(\alpha_n)$) but needs to be a Martingale difference sequence. The noise e_2 is smaller (of $\mathcal{O}(\alpha_n^2)$) but does not need to have other special properties. We will shortly see how Markovian samples can fit into the two noise terms but Theorem 4.7 is already enough for the almost sure convergence of (Linear TD) and (Q-learning) with i.i.d. samples. Specifically, if $\{Y_t\}$ is i.i.d. in (2), we can

identify $e_{1,n+1}$ as $\alpha_t(F(w_t, Y_{t+1}) - f(w_t))$ and $e_{2,n+1}$ as 0 and then invoke Theorem 4.7. To work with Markovian $\{Y_t\}$, we follow the skeleton iterates technique in Qian et al. (2024), which is essentially an improved version of a proof technique used in the proof of Proposition 4.8 of Bertsekas & Tsitsiklis (1996). We use $G(w, Y) \doteq F(w, Y) - w$ and $g(w) \doteq f(w) - w$ as shorthands. We then consider a deterministic and strictly increasing sequence $\{t_m\}_{m=0,1,...}$, called the anchors, with $t_0 = 0$ and $\lim_{m\to\infty} t_m = \infty$. Telescoping (1) then yields that for any m,

$$\begin{aligned} w_{t_{m+1}} =& w_{t_m} + \textstyle\sum_{t=t_m}^{t_{m+1}-1} \alpha_t G(w_t, Y_{t+1}) \\ =& w_{t_m} + \beta_m g(w_{t_m}) + \textstyle\sum_{t=t_m}^{t_{m+1}-1} \alpha_t (G(w_{t_m}, Y_{t+1}) - \mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}]) \\ &+ \textstyle\sum_{t=t_m}^{t_{m+1}-1} \alpha_t (\mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}] - g(w_{t_m}) + G(w_t, Y_{t+1}) - G(w_{t_m}, Y_{t+1})), \end{aligned}$$

where $\beta_m \doteq \sum_{t=t_m}^{t_{m+1}-1} \alpha_t$. We can then in (3) identify $x_n(\omega)$ as w_{t_m}, α_n as β_m, $\mathcal{F}_n$ as $\mathcal{F}_{t_m}$, $e_{1,n+1}(\omega)$ as $\sum_{t=t_m}^{t_{m+1}-1} \alpha_t (G(w_{t_m}, Y_{t+1}) - \mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}])$, and $e_{2,n+1}(\omega)$ as $\sum_{t=t_m}^{t_{m+1}-1} \alpha_t (\mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}] - g(w_{t_m}) + G(w_t, Y_{t+1}) - G(w_{t_m}, Y_{t+1}))$. Verifying Assumptions 4.1 - 4.5 and invoking Theorem 4.7 then concludes that $\lim_{m\to\infty} w_{t_m} = w_*$ a.s. A few Gronwall's inequalities further give $\lim_{t\to\infty} w_t = w_*$. See Qian et al. (2024) for a detailed proof in natural language. Notably, the anchors $\{t_m\}$ need an additional property that $\alpha_{t_m} \le C\beta_m^2$. For this to hold, we need $\nu \in (2/3, 1)$ in Theorems 3.1 & 3.2. For $\nu = 1$, Qian et al. (2024) have already also proved it. So it will be straightforward to formalize. For $\nu \in (1/2, 2/3]$, we need to either extend the results of Qian et al. (2024) or resort to the canonical ODE based approach (Benveniste et al., 1990; Kushner & Yin, 2003; Borkar, 2009; Borkar et al., 2025; Liu et al., 2025a), among which our evaluation is that Liu et al. (2025a) is the most plausible to formalize and is perhaps the most powerful in terms of almost sure convergence. For $\nu \in (0, 1/2]$, the $\{\alpha_n\}$ even does not satisfy the Robbins-Monro condition and we need the more recent ODE approach from Lauand & Meyn (2024).

5 Conclusion

This paper provides the first formalization of the almost sure convergence of linear TD and Q-learning, significantly advancing the state of the art in formalizing RL theory. The developed framework is immediately ready to formalize more convergent RL results. By taking conditional expectations on both sides of (5) and telescoping, we can immediately get convergence rates in $\mathcal{L}_2$ with i.i.d. samples. To get $\mathcal{L}_2$ convergence rates with Markovian samples, one can apply the technique from Srikant & Ying (2019). Following Qian et al. (2024); Liu et al. (2026b), we can also obtain almost sure convergence rates easily under Markovian samples, after getting a nonasymptotic version of the Robbins-Siegmund theorem following Liu & Yuan (2022); Karandikar & Vidyasagar (2024). We believe the aforementioned formalization shall be straightforward. Next is concentration with exponential tails and $\mathcal{L}_p$ convergence. For this, we will need techniques from Chen et al. (2025) for i.i.d. samples and Qian et al. (2024) for Markovian samples. Both should be straightforward if we can formalize Hoeffding's lemma. Our framework can be extended to other off-policy TD methods as well. The family of gradient TD methods (Sutton et al., 2008; 2009; Yu, 2017; Zhang et al., 2021; Qian & Zhang, 2025) is straightforward to formalize, suppose no eligibility trace is involved. The family of emphatic TD methods (Yu, 2015; Sutton et al., 2016) is much harder unless the trace is truncated (Zhang & Whiteson, 2022) as the full trace will inevitably incur analysis of very complicated general state space Markov chains. More challenging are the algorithms involving time-inhomogeneous Markov chains, e.g., (linear) SARSA (Zou et al., 2019), (linear) Q-learning with a changing behavior policy (Meyn, 2024; Liu et al., 2025c;b), and policy gradient methods (Sutton et al., 1999; Konda, 2002; Agarwal et al., 2020; Mei et al., 2020; Zhang et al., 2022). For those algorithms, the Markov chain is deeply coupled with the iterates and we envision major updates of our framework are necessary.

The most technically challenging part of this project comes from conditional expectation. For example, consider computing $\mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}]$ where $\{Y_t\}$ is a finite Markov chain and we use a matrix $P_{\mathcal{Y}}$ to denote its transition kernel. It is straightforward for humans to conclude that

$$\mathbb{E}[G(w_{t_m}, Y_{t+1})|\mathcal{F}_{t_m}] = \textstyle\sum_y P_{\mathcal{Y}}^{t+1-t_m}(Y_{t_m}, y) G(w_{t_m}, y). \tag{6}$$

But to formalize this in Lean is highly nontrivial. The difficulties come from two aspects. First, the conditional expectation in Lean is defined in an abstract and measure-theoretic way. So many intuitive results about conditional expectation are highly nontrivial to formalize. Second, the probability space used for this measure-theoretic definition of conditional expectation is generated by the Ionescu-Tulcea theorem, which is formalized in Lean in a constructive way for a generic family of history-dependent kernels. So here we have to go into the details of the construction and simplify it for a Markov chain. Formalizing (6) takes roughly 1,000 lines of Lean code, about 10% of the entire project. As a reference, Sonoda et al. (2025) explicitly state that they use a customized proof in FoML to entirely avoid conditional expectation. Addressing such small last-mile friction in formalization is a promising direction for future work. And recent works such as Liu et al. (2026a); Xie et al. (2026) have made some initial progress in this direction.

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Supplementary Materials

The following content was not necessarily subject to peer review.

A Lean and Mathlib in a Nutshell

Lean is an interactive theorem prover based on dependent type theory: every theorem is a type, and a proof is a term inhabiting that type (de Moura & Ullrich, 2021). Lean's trusted kernel checks proofs by typechecking, so correctness reduces to a small, auditable code base. In practice, users rarely write raw proof terms. Instead, Lean provides a tactic framework that lets users build proofs step-by-step using high level commands (e.g., `simp`, `rw`, `apply`). Mathlib is Lean's flagship community-maintained library of formalized mathematics (The mathlib Community, 2020). It supplies definitions, theorems, and a large ecosystem of automation and conventions. For readers unfamiliar with proof assistants, the key takeaway is: Lean is reliable because the kernel checks every detail, but writing a formal proof in Lean that goes through depends heavily on handling all the details in the proof carefully, even if they may look trivial to humans.

It is useful to separate two workflows in Lean. *(1) Formalizing existing statements:* starting from an informal theorem statement in a paper or textbook and producing a well-typed Lean statement (without proof) that captures the intended meaning using Mathlib's definitions and conventions. This step requires many design choices (which definitions to use, which generality and typeclass assumptions are idiomatic, which normal forms are preferred). In general, there is no way to automatically check whether the formalization is correct and we have to resort to human experts. *(2) Proving a Lean statement:* given a type-checked Lean statement, producing a Lean proof that the kernel accepts. The correctness can be automatically checked by Lean compiler, which is the major advantage of formalized mathematics. This work touches (1) and (2) in Section 3 and Section 4, respectively.

B Proof of (4)

$$
\begin{aligned}
&\left\langle \nabla \tfrac{1}{2}\|q-q_*\|_p^2, f(q)-q \right\rangle \\
=&\left\langle \nabla \tfrac{1}{2}\|q-q_*\|_p^2, \mathcal{T}_*'q-q_* \right\rangle + \left\langle \nabla \tfrac{1}{2}\|q-q_*\|_p^2, q_*-q \right\rangle \\
=&\left\langle \nabla \tfrac{1}{2}\|q-q_*\|_p^2, \mathcal{T}_*'q-q_* \right\rangle - \|q-q_*\|_p^2 && \text{(By computation)} \\
\le&\left\| \nabla \tfrac{1}{2}\|q-q_*\|_p^2 \right\|_{(1-p^{-1})^{-1}} \|\mathcal{T}_*'q-q_*\|_p - \|q-q_*\|_p^2 && \text{(By Holder's inequality)} \\
=&\|q-q_*\|_p \|\mathcal{T}_*'q-q_*\|_p - \|q-q_*\|_p^2 && \text{(By computation)} \\
\le&(|\mathcal{S}||\mathcal{A}|)^{1/p}\|q-q_*\|_p \|\mathcal{T}_*'q-q_*\|_\infty - \|q-q_*\|_p^2 && \text{(By norm equivalence)} \\
\le&\gamma'(|\mathcal{S}||\mathcal{A}|)^{1/p}\|q-q_*\|_p \|q-q_*\|_\infty - \|q-q_*\|_p^2 \\
\le&-(1-\gamma'(|\mathcal{S}||\mathcal{A}|)^{1/p})\|q-q_*\|_p^2. && \text{(By norm equivalence)}
\end{aligned}
$$